PolyFlop8

ISTQB Test Requirements Specification

Block Level & ISA Execution Verification

Revision: 1.2
Date: January 5, 2026

Based on PolyFlop8 Technical Reference Manual

1 HARDWARE BLOCK TEST REQUIREMENTS

This section defines the functional requirements for individual hardware components (Unit Testing).

1.1 ALU (Arithmetic Logic Unit)

Req ID	Requirement Description	Traceability
TR-ALU-01	Verify all arithmetic/logical operations (ADD, SUB, AND, XOR, etc.) defined in the Operations Table produce correct 8-bit results.	Table 2.3
TR-ALU-02	Verify flags (H, V, N, C, Z) are correctly asserted for each operation type according to the "Flags Affected" column.	Sec 2.2
TR-ALU-03	Verify `alu_mux_sel_a` correctly selects between `regfile_data_a` ('0') and `pc_data` ('1').	Sec 2.2
TR-ALU-04	Verify `alu_mux_sel_b` correctly selects between `regfile_data_b` ('0') and `rs_imm_in` ('1').	Sec 2.2
TR-ALU-05	Verify 2's complement handling in NEG and SUB operations, specifically checking Overflow (V) flag logic.	Table 2.3

1.2 Instruction Decoder

Req ID	Requirement Description	Traceability
TR-DEC-01	Verify `opcode` output correctly extracts bits [15:11] from the 16-bit `decoder_data_in`.	Table 2, Slice 1
TR-DEC-02	Verify `RD` output correctly extracts bits [10:8] representing the destination register.	Table 2, Slice 2
TR-DEC-03	Verify `RS_IMM` output correctly extracts bits [7:0] for use as either Source Register or Immediate value.	Table 2, Slice 3
TR-DEC-04	Verify `Jump11bit` output correctly extracts bits [10:0] for absolute jump addresses.	Table 2, Slice 4

1.3 Program Counter (PC)

Req ID	Requirement Description	Traceability
TR-PC-01	Verify **Fetch Mode** (`pc_src="00"`): PC increments by 1 on each clock cycle.	Table 4.3
TR-PC-02	Verify **Branch Mode** (`pc_src="01"`): PC updates to `(PC + 1) + sign_extend(alu_out)`.	Table 4.3
TR-PC-03	Verify **Indirect/Ret Mode** (`pc_src="10"`): PC loads address from `ram_data`.	Table 4.3
TR-PC-04	Verify **Jump Mode** (`pc_src="11"`): PC loads absolute address from `jump_abs_11bit`.	Table 4.3
TR-PC-05	Verify `pc_en` signal effectively stalls the PC (no change) when low.	Sec 4.2

1.4 Register File

Req ID	Requirement Description	Traceability
TR-REG-01	Verify asynchronous read on `data_a` and `data_b` based on addresses `addr_a` and `addr_b`.	Sec 5
TR-REG-02	Verify synchronous write to register at `addr_w` only when `we` is high.	Sec 5.3
TR-REG-03	Verify `data_r7x` acts as a dedicated hardwired output for Register 7.	Sec 5.2
TR-REG-04	Verify Write Mux (`rf_src_sel`) correctly selects sources: ALU, RAM, Immediate, or I/O.	Sec 5.2

1.5 Data RAM

Req ID	Requirement Description	Traceability
TR-RAM-01	Verify asynchronous read returns the value stored at the selected address without requiring a clock edge.	DataRAM, Read Logic
TR-RAM-02	Verify synchronous write updates memory only on rising clock edge when `mem_we = '1'`.	DataRAM, Write Process
TR-RAM-03	Verify Address MUX selects correct source based on `mem_addr_sel`: ALU, R7, Reg A, or Immediate.	DataRAM Addr MUX
TR-RAM-04	Verify Data MUX selects correct write data source: Reg B, PC, or ALU output.	DataRAM Data MUX
TR-RAM-05	Verify memory boundary behavior for lowest (0x00) and highest (0xFF) addresses.	DataRAM Address Range

1.6 Status Register (SREG)

Req ID	Requirement Description	Traceability
TR-SREG-01	Verify flags H, V, N, Z, C are correctly latched from ALU outputs when `sreg_we = '1'` and `sreg_src = '0'`.	StatusReg ALU Mode
TR-SREG-02	Verify full 8-bit SREG is loaded from RegFile when `sreg_src = '1'` (POP/OUT behavior).	StatusReg RegFile Mode
TR-SREG-03	Verify Sign flag (S) is computed as `N xor V`.	StatusReg Bit Mapping
TR-SREG-04	Verify bits I and T retain previous values during ALU flag updates.	StatusReg Latch Logic
TR-SREG-05	Verify asynchronous reset clears entire Status Register.	StatusReg Reset

1.7 Program ROM (Instruction Memory)

Req ID	Requirement Description	Traceability
TR-ROM-01	Verify instruction fetch returns correct 16-bit word for given 11-bit PC address.	ProgPROM Read Logic
TR-ROM-02	Verify ROM read is asynchronous and does not depend on clock edge.	ProgPROM Architecture
TR-ROM-03	Verify correct instruction encoding for LDI, ADD, SUB, and JMP instructions.	ROM Program Image
TR-ROM-04	Verify JMP instruction correctly maps 11-bit address field into instruction word.	ProgPROM Opcode Format
TR-ROM-05	Verify undefined addresses return NOP-equivalent instruction.	ROM Default Case

1.8 I/O Unit

Req ID	Requirement Description	Traceability
TR-IO-01	Verify OUT instruction writes register value to output port when `io_we = '1'`.	IO_Unit Write Logic
TR-IO-02	Verify correct output port selection based on `io_addr_in`.	IO Address Decode
TR-IO-03	Verify IN instruction reads external pin values into register via `io_data_in`.	IO Read Logic
TR-IO-04	Verify output port retains last written value until updated or reset.	IO Output Register
TR-IO-05	Verify reset clears all output port registers to zero.	IO Reset Logic

1.9 Control Unit

Req ID	Requirement Description	Traceability
TR-CU-01	Verify FSM transitions correctly through FETCH $\rightarrow$ DECODE $\rightarrow$ EXECUTE states.	ControlUnit FSM
TR-CU-02	Verify instruction fetch enables IR write only during FETCH state.	IR Enable Logic
TR-CU-03	Verify correct ALU control signals are generated for arithmetic instructions.	ALU Control Mapping
TR-CU-04	Verify memory write enable is asserted only for STORE instructions.	DRAM Control
TR-CU-05	Verify register file write-back occurs only for instructions producing results.	RegFile WE Logic
TR-CU-06	Verify PC update source is correctly selected for sequential, branch, and jump instructions.	PC Control Logic
TR-CU-07	Verify default control outputs prevent unintended writes (NOP safety).	Default Signal Values

2 ISA EXECUTION TEST REQUIREMENTS

This section defines requirements for System Integration Testing, verifying the CPU executes code correctly (Equivalence Partitioning & Boundary Value Analysis).

2.1 Arithmetic & Logic Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-01	Verify ADD and ADC correctly sum two 8-bit registers (Equivalence Partitioning).	ADD, ADC
TR-ISA-02	Verify ADD sets the **Carry Flag (C)** when result exceeds 255 (Boundary Value).	ADD
TR-ISA-03	Verify ADD sets the **Overflow Flag (V)** when signed 2's complement addition wraps incorrectly.	ADD
TR-ISA-04	Verify SUB and SBB correctly subtract two 8-bit registers (checking Reg-Reg and Reg-Imm).	SUB, SBB
TR-ISA-05	Verify SUB sets the **Zero Flag (Z)** only when the result is exactly 0.	SUB, CMP
TR-ISA-06	Verify Logical Operations (AND, OR, XOR) correctly manipulate bits and clear/set flags.	AND, OR
TR-ISA-07	Verify INC and DEC increment/decrement target without affecting Carry flag (if specified).	INC, DEC

2.2 Data Transfer Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-08	Verify **Immediate Load:** Loading a constant (e.g., LDI R1, 0xFF) results in exact value in destination.	LDI
TR-ISA-09	Verify **Register Move:** Copying data from Rs to Rd preserves value; Source unchanged.	MOV
TR-ISA-10	Verify **Direct Store:** Storing register to memory address (ST [Addr], R1) writes correct data.	STS
TR-ISA-11	Verify **Direct Load:** Loading from a valid memory address retrieves previously written data.	LDS
TR-ISA-12	Verify **Indirect Addressing:** Accessing memory via pointer (LD R1, [R7]) accesses address in R7.	LD/ST

2.3 Program Flow Control

Req ID	Test Requirement Description	Instruction
TR-ISA-13	Verify **Unconditional Jump:** JMP updates PC to target immediately, regardless of flags.	RJMP
TR-ISA-14	Verify **Conditional Branch (Taken):** BR updates PC only if condition (e.g., Z=1) is met.	BRBS/BRBC
TR-ISA-15	Verify **Conditional Branch (Not Taken):** BR proceeds to PC + 1 if condition is NOT met.	BRBS/BRBC
TR-ISA-16	Verify **Call:** RCALL pushes Return Address (PC + 1) onto Stack before jumping.	RCALL
TR-ISA-17	Verify **Return:** RET pops top value from Stack into PC, resuming execution.	RET

2.4 Stack & I/O Instructions

Req ID	Test Requirement Description	Instruction
TR-ISA-18	Verify **PUSH/POP:** Pushing/Popping correctly updates Stack Pointer and memory.	PUSH/POP
TR-ISA-19	Verify **I/O OUT:** OUT instruction writes register value to specified Output Port latch.	OUT
TR-ISA-20	Verify **I/O IN:** IN instruction reads current logic level of Input Port pins into register.	IN